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Controlling Protein Transport in Ultrafiltration Using Small Charged Ligands

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Abstract: Previous studies have demonstrated that protein transport during ultrafiltration can be strongly influenced by solution pH and ionic strength. The objective of this study was to examine the possibility of controlling protein transmission using a small, highly charged ligand that selectively binds to the protein of interest. Experiments were performed using bovine serum albumin and the dye Cibacron Blue. Protein sieving data were obtained with essentially neutral and negatively charged versions of a composite regenerated cellulose membrane to examine the effects of electrostatic interactions. The addition of only 1 g/L of Cibacron Blue to an 8 g/L BSA solution reduced the BSA sieving coefficient through the negatively-charged membrane by more than two orders of magnitude, with this effect being largely eliminated at high salt and with the neutral membrane. Protein sieving data were in good agreement with model calculations based on the partitioning of a charged sphere in a charged pore accounting for the change in net protein charge due to ligand binding and the increase in solution ionic strength due to the free ligand in solution.

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INTRODUCTION

A number of recent studies have demonstrated that it is possible to control the rate of protein transport through ultrafiltration membranes by adjusting the solution pH or the ionic strength. For example Pujar and Zydney (1994) showed that the transmission of bovine serum albumin (BSA) through a 100,000 molecular weight cut-off membrane could be reduced by a factor of 100 by simply lowering the solution ionic strength from 150 to 1.5 mM. This dramatic increase in protein retention was a direct result of the electrostatic exclusion of the negatively charged protein from the membrane pores at low salt concentrations. Burns and Zydney (1999) demonstrated that the sieving coefficient of several therapeutic proteins could be reduced by an order of magnitude by shifting the solution pH approximately 2 pH units above or below the protein isoelectric point due to the increase in the effective protein volume associated with the

presence of the electrical double layer surrounding the charged protein.

These electrostatic interactions can also be exploited for protein separations using a process known as High Performance Tangential Flow Filtration or HPTFF. For example, van Reis et al. (1999) were able to achieve 900-fold purification of an antigen-binding fragment of a monoclonal antibody (Fab) from BSA by operating the membrane process near the isoelectric point of the BSA and using a positively-charged membrane to obtain very high rejection of the positively-charged Fab.

Although the use of solution pH and ionic strength to control protein transmission is very attractive, there are many systems where this is not practical due to issues of protein stability or when the product and impurity have very similar surface charge characteristics. Menon and Zydney (1999) have demonstrated that it is also possible to control protein transmission through the choice of buffer due to the change in protein charge caused by ion binding. For example, the BSA sieving coefficient at pH 4.4 was 50% larger in a 10 mM solution of NaSCN than in a corresponding solution of NaCl due to the differences in anion binding, although this effect was relatively small under most conditions and ion binding interactions typically have low specificity. Kanani et al. (2004) showed that the addition of BSA increased the retention of lysozyme, which was attributed to the attractive interactions between lysozyme and the large molecules of BSA. In contrast, the presence of BSA reduced the transmission of myoglobin, which Kanani et al. attributed to Donnan effects. The combination of these phenomena actually reversed the selectivity for the separation of lysozyme and myoglobin. However, it is unlikely that this general approach would be practical for commercial separations given the high cost of the added protein and the need to subsequently remove that protein to parts per million levels.

A much more attractive approach would be to use a small charged ligand that binds specifically to the protein of interest, thereby altering the protein charge and in turn its transmission during ultrafiltration. Charged dyes have been used extensively in dye–ligand affinity chromatography; thus, there is a considerable database already available for

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dye–protein interactions that can be used to identify appropriate candidate structures. For example, Cibacron Blue 3GA is known to bind to a variety of proteins including many dehydrogenases and kinases, nucleotide dependent enzymes, reductases, interferons, and serum protein such as albumin (Lowe, 1987). The versatility of Cibacron Blue binding is apparently due to the combination of aromatic and multiple sulphonate groups within the dye molecule (Subramanian, 1984). Cibacron Blue has been shown to bind to BSA at multiple hydrophobic fatty acid-anion binding sites on the protein surface (Leatherbarrow and Dean, 1980).

The overall objective of this study was to determine the effect of Cibacron Blue on the ultrafiltration of BSA, with the ultimate goal of being able to use this small dye to control BSA transmission through the membrane. Experiments were performed with essentially neutral and negatively-charged versions of a composite regenerated cellulose membrane to explore the effects of electrostatic interactions on protein transmission. The data were analyzed using available theoretical models for the transport of charged proteins through charged membranes, with the protein charge calculated by accounting for the extent of dye binding.

MATERIALS AND METHODS

Experiments were performed using Bovine Serum Albumin (Fraction V heat shock precipitated BSA, catalogue # A 7906 from Sigma Chemical, St. Louis, MO). BSA has an isoelectric point (pI) of approximately 4.9 and a molecular weight of 69 kDa. Cibacron Blue 3GA (Catalogue # C-9534, Sigma Chemical), which has a molecular weight of 0.774 kDa and contains three negatively-charged sulphonic acid groups, was used as the small affinity ligand due to its extensive binding interactions with BSA.

BSA solutions were prepared using an acetate buffer, which was made by dissolving pre-weighed amounts of CH_3COONa and CH_3COOH (EM Science, Gibbstown, NJ) in deionized distilled water obtained from a NANOpure Diamond water purification system (Barnstead Thermolyne Corporation, Dubuque, IA) with a resistivity greater than 18 $\text{M}\Omega\text{-cm}$. The ionic strength of the buffer solution was adjusted by adding appropriate amounts of NaCl (Mallinckrodt Baker, Inc., Phillipsburg, NJ). The solution pH was measured using a 420APlus pH meter (Thermo Orion, Beverly, MA). All solutions were prefiltered through 0.2 μm pore size Super-200 membranes (Pall Corp., Ann Arbor, MI) to remove particulates. Protein solutions were prepared by slowly dissolving BSA powder in the buffer, with the resulting solution filtered through a 0.22 μm syringe filter (Costar Corp., Cambridge, MA) to remove any protein aggregates immediately prior to use.

Protein concentrations in the absence of the dye were determined spectrophotometrically with a UV-VIS spectrophotometer (UV mini 1240, Shimadzu, Kyoto, Japan) using the natural absorbance at 280 nm. Actual concentrations were evaluated by comparison of the measured absorbance with that of standard solutions of known concentrations.

Protein concentrations in the presence of Cibacron Blue were evaluated by capillary zone electrophoresis (Model G1600A, Agilent Technologies, Palo Alto, CA) since the strong blue color of Cibacron Blue interfered with the absorbance measurements. Data were obtained using fused silica capillaries with an inner diameter of 50 μm and an effective length of 72 cm (Catalogue # G 1600–62211, Agilent Technologies). The applied voltage was maintained at 25 kV, and a 20 mM borate buffer at pH 9.3 (Catalogue # 8500–6782, Agilent Technologies, Waldbronn, Germany) with 10% acetonitrile (EM Science) on a volume/volume basis was used as the background electrolyte. Protein detection was at a wavelength of 214 nm, with the protein concentration determined by numerical integration of the area under the corresponding peak. Actual concentrations were evaluated using previously constructed calibration curves determined for BSA in the presence of Cibacron Blue.

Cibacron Blue concentrations were determined spectrophotometrically using the natural absorbance at 616 nm. The presence of BSA had no effect on the absorbance of the dye at 616 nm over the range of concentrations examined in this study.

All filtration experiments were performed with Composite Regenerated Cellulose (CRC) membranes (Millipore Corp., Bedford, MA) having nominal molecular weight cut offs of 30 or 100 kDa. These cellulosic membranes are nearly uncharged and have very low protein adsorption due to their high degree of hydrophilicity. Individual membrane discs (25 mm diameter) were cut from large flat sheets using a specially designed cutting device.

A negatively-charged version of the 100 kDa membrane was made in our laboratory by the covalent attachment of negatively charged sulphonic acid groups to the surface of the membrane using the base-activated chemistry developed by van Reis (2001). Membranes were first flushed with deionized distilled water to remove any residual storage agents. The membranes were then equilibrated with 0.1 N NaOH and immersed in a 2 M solution of 3-Bromopropylsulfonic acid sodium salt (Catalogue #B2912, Sigma Chemical) in 0.1 N NaOH for approximately 24 h. The membranes were then flushed with approximately 100 L/m^2 of 0.1 N NaOH followed by 100 L/m^2 distilled deionized water.

Equilibrium Binding Experiments

The binding interactions between BSA and Cibacron Blue were evaluated using a stirred ultrafiltration cell which allowed protein-free samples to be collected through the UF membrane. A BSA–Cibacron Blue mixture was added to a 25 mm diameter stirred UF cell (Model 8010, Amicon Corp., Beverly, MA) which contained a 30 kDa nominal molecular weight cut-off CRC membrane that is fully retentive to BSA. The stirrer was set at 600 rpm using a Strobotac Type 1531-AB strobe light (General Radio Co., Concord, MA). The device was then air pressurized to $\Delta P = 21 \text{ kPa}$ (3 psig), and small samples were withdrawn through the membrane for

off-line analysis of the free Cibacron Blue concentration. Data were also obtained with pure Cibacron Blue (in the absence of BSA) to evaluate the sieving coefficient of free Cibacron Blue through the 30 kDa membrane.

Protein Filtration

Protein filtration experiments were also conducted in the 25 mm diameter stirred ultrafiltration cell. The stirred cell was initially filled with acetate buffer taking care to remove all trapped air bubbles from the cell and tubing. The stirrer speed was set to 600 rpm using a Strobotac Type 1531-AB strobe light, and the device was air pressurized to the desired value. The membrane hydraulic permeability (L_p) was evaluated from the slope of data for the filtration velocity (J_v) as a function of the transmembrane pressure drop (ΔP):

$$L_p = \frac{J_v \mu}{\Delta P} \quad (1)$$

where μ is the solution viscosity. The stirred cell was then emptied and refilled with a BSA-Cibacron Blue solution at a desired ionic strength and pH. The stirrer was reset, the cell was air-pressurized, and the filtrate flux was determined by timed collection using a digital balance (Model AG104, Mettler Toledo, Columbus, OH) with an accuracy of 0.1 mg. A minimum of 500 μ L of filtrate was flushed through the membrane to ensure washout of the dead volume beneath the membrane at the bottom of the stirred cell. A 200 μ L filtrate sample was then collected for subsequent analysis of the BSA and Cibacron Blue concentrations. The filtrate port was immediately clamped and a small (approximately 200 μ L) sample of the bulk solution was obtained directly from the stirred cell. The stirred cell was then emptied, rinsed, and refilled with acetate buffer, with the hydraulic permeability of the used membrane evaluated to provide a measure of the extent of fouling. All experiments were performed at room temperature ($22 \pm 3^\circ\text{C}$) using the same lot of protein.

RESULTS AND ANALYSIS

Binding Isotherms

Before examining the effects of Cibacron Blue on protein sieving, we first evaluated the extent of Cibacron Blue binding to BSA, with the results at pH 5.0, 10 mM ionic strength, and a total BSA concentration of 3 g/L shown in Figure 1. The x-axis represents the concentration of free Cibacron Blue, which was evaluated directly from the measured filtrate concentrations obtained through the 30 kDa membrane as:

$$C_{\text{free}} = \frac{C_{\text{filtrate}}}{S_0} \quad (2)$$

The observed sieving coefficient for free Cibacron Blue (S_0) was evaluated immediately prior to performing the binding experiment using the same concentration of pure Cibacron Blue that was added to the protein solution. We assumed

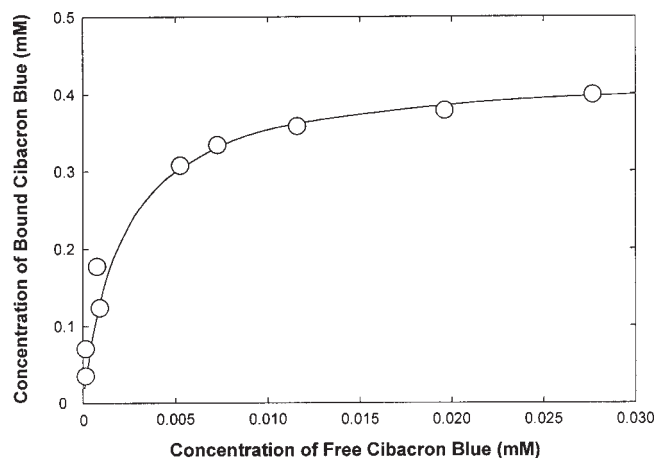


Figure 1. Binding isotherm for Cibacron Blue at pH 5 in a 10 mM acetate buffer at a BSA concentration of 0.043 mM (3 g/L). Solid curve represents model calculation using binding parameters in Table I.

that the transmission of free (unbound) Cibacron Blue was unaffected by the presence of the protein; this behavior was confirmed experimentally from limited data obtained with a neutral CRC 100 kDa membrane that was non-retentive to the free dye but slightly permeable to both BSA and the protein-dye complex. The sieving coefficient of free Cibacron Blue varied from 0.53 for a 0.28 g/L solution to 0.80 for a 8.3 g/L solution, all in 10 mM acetate buffer at pH 5.0. The small degree of retention of Cibacron Blue was due to the electrostatic repulsion between the highly charged dye and the small number of negatively-charged groups on the cellulose membrane, which has a zeta potential of approximately -0.0002 C/m^2 under the conditions of these experiments. The zeta potential was determined from streaming potential experiments as described by Burns and Zydney (2000). The extent of electrostatic exclusion decreases as the Cibacron Blue concentration increases due to the corresponding increase in the solution ionic strength. This phenomenon is discussed in more detail by Shao and Zydney (2004). There was no evidence of any leakage of the 69 kDa BSA through the 30 kDa membrane, as determined by capillary electrophoresis of the filtrate samples, for any of the experimental conditions examined in this study.

The amount of bound Cibacron Blue was determined from a simple mass balance:

$$C_{\text{bound}} = C_{\text{total}} - C_{\text{free}} \quad (3)$$

where the total concentration of Cibacron Blue in the solution was evaluated directly from the mass of dye added to the known volume of buffer. The concentration of bound Cibacron Blue rapidly increases with increasing concentration of free Cibacron Blue, but then appears to approach an asymptote of approximately 0.41 mM at high Cibacron Blue concentrations. The maximum concentration of bound Cibacron Blue is approximately 9.5 times larger than the concentration of BSA in this experiment (3 g/L = 0.043 mM), which is consistent with the presence

of multiple binding sites for Cibacron Blue on each molecule of BSA (Antoni et al., 1982; Johansson and Joelsson, 1991).

Previous studies have shown that Cibacron Blue binds to BSA only at fatty acid binding sites (Leatherbarrow and Dean, 1980), suggesting that the binding data might be effectively fit using a simple multisite Langmuir-type binding isotherm of the form:

$$C_{\text{bound}} = \frac{nK_{\text{eq}}C_{\text{BSA}}C_{\text{free}}}{1 + K_{\text{eq}}C_{\text{free}}} \quad (4)$$

where n is the number of moles of Cibacron Blue bound per mole of BSA, K_{eq} is the equilibrium binding constant, and C_{BSA} is the total concentration of BSA in the solution. The binding data from Figure 1 have been replotted in Figure 2 in the form of a Hanes plot, in which the ratio $C_{\text{free}}/C_{\text{bound}}$ is plotted as a function of C_{free} . The data are highly linear when plotted in this manner, with R^2 values greater than 0.99 for all experiments. The Hanes plot was used to obtain initial estimates of the binding parameters, with slope $1/nC_{\text{BSA}} = 2.46 \text{ mM}^{-1}$ and intercept $1/(nK_{\text{eq}}C_{\text{BSA}}) = 0.003$.

More accurate values of the binding parameters were obtained by fitting the experimental data over a range of BSA concentrations from 1.5 to 5 g/L using a non-linear least squares (NNLS) Marquardt-Levenberg algorithm, with the individual data points weighted by the inverse of the standard error for each measurement. The best fit values for n and K_{eq} are shown in Table I at three different ionic strengths, with the model calculations shown as the solid lines in Figures 1 and 2. The large error bars on the equilibrium constants, particularly in the high ionic strength solution, are due to the very strong binding between BSA and Cibacron Blue, which made it very difficult to obtain significant data in the linear region of the isotherm. The maximum binding level decreased slightly with increasing ionic strength, varying from 9.8 ± 0.2 in the 10 mM solution to 8.8 ± 0.3 in the 100 mM solution. These values are slightly larger than the 7.5 moles of

Table I. Equilibrium parameters for Cibacron Blue binding to BSA.

Ionic strength (mM)	Number of binding sites, n	Equilibrium constant, K_{eq} (10^5 M^{-1})
10	9.8 ± 0.2	4.7 ± 1.2
50	9.5 ± 0.9	0.2 ± 0.1
100	8.8 ± 0.3	0.3 ± 0.08

Cibacron Blue bound per mole of human serum albumin (HSA) reported by Compagnini et al. (1996) for data obtained in a 25 mM sodium acetate, 100 mM NaCl, pH 4.5 buffer, with these differences probably due to differences in experimental conditions and in the detailed physicochemical characteristics of the BSA and HSA (possibly including differences in the amount of bound fatty acid).

The equilibrium binding constant was greatest in the 10 mM ionic strength solution with a value of $4.7 \pm 1.2 \times 10^5 \text{ M}^{-1}$ compared to only 0.2 or $0.3 \times 10^5 \text{ M}^{-1}$ at the higher salt concentrations. The larger value of K_{eq} in the 10 mM solution may be due to the greater attractive electrostatic interactions between the negatively charged Cibacron Blue and the large number of positively charged lysine amino groups on BSA. Literature values for K_{eq} are highly scattered; Antoni et al. (1982) reported $K_{\text{eq}} = 10^6 \text{ M}^{-1}$ in a 0.05 M tris-HCl, 0.5 M NaCl, pH 8.0 buffer while Metcalf et al. (1981) found a value of $K_{\text{eq}} = 5.1 \times 10^3 \text{ M}^{-1}$ in a column equilibrated with 25 mM tris-HCl, pH 8.0 buffer. The values obtained in this study fall within this broad range.

Protein Sieving

Figure 3 shows experimental data for the observed sieving coefficient of BSA through a negatively-charged version of the 100 kDa CRC membrane as a function of the total concentration of Cibacron Blue added to the solution. The data were all obtained using a BSA concentration of 8 g/L in a 10 mM acetate buffer at pH 5.0. In each case, the observed sieving coefficient (S_o) was evaluated as the ratio of the

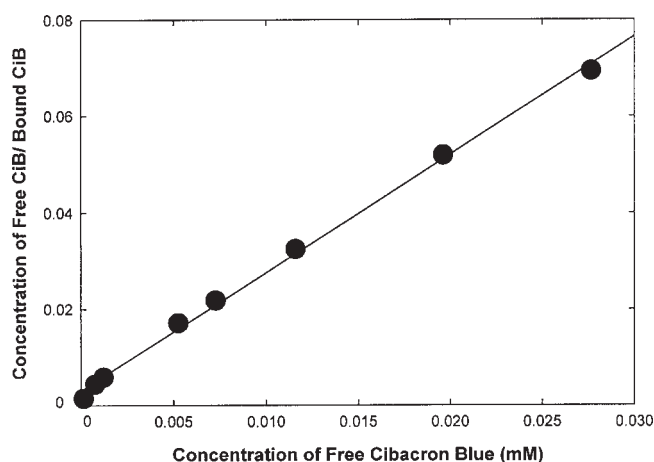


Figure 2. Hanes plot of the data in Figure 1 showing linearized form of the binding isotherm.

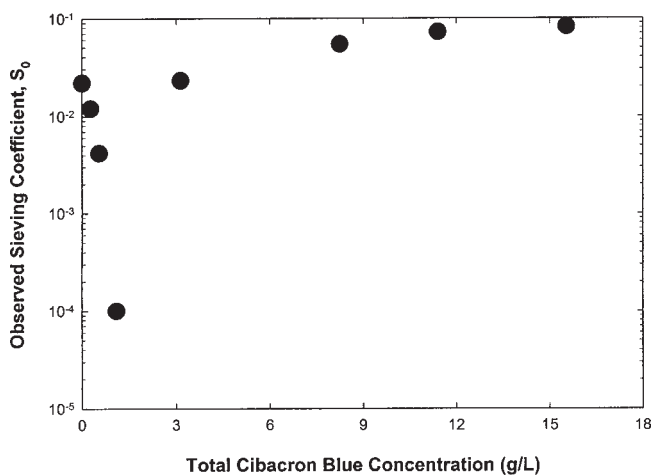


Figure 3. Observed BSA sieving coefficients as a function of total Cibacron Blue concentration. Data were obtained using a negatively charged 100 kDa CRC membrane in a 10 mM acetate buffer at pH 5.0 with 8 g/L BSA.

BSA concentration in the filtrate solution to that in the bulk solution in the stirred cell, with the BSA concentrations determined by capillary electrophoresis. The BSA sieving coefficient initially decreases with increasing concentration of Cibacron Blue, with S_o decreasing from a value of 0.02 in the absence of Cibacron Blue to a value of 0.0001 at a Cibacron Blue concentration of approximately 1.1 g/L. This very low sieving coefficient was at the detection limit for the assay used in this study. Further increases in the Cibacron Blue concentration cause an increase in BSA transmission, with the sieving coefficient attaining a value of more than 0.08 at a Cibacron Blue concentration above 15 g/L. Thus, the addition of several g/L of Cibacron Blue causes the BSA sieving coefficient to vary by more than a factor of 800 fold.

The location of the minimum in the BSA sieving coefficient seen in Figure 3 occurs at a molar ratio of Cibacron Blue to BSA of approximately 12, which is only slightly larger than the maximum number of molecules of Cibacron Blue that can be bound to a molecule of BSA under these experimental conditions ($n=9.8$ in the 10 mM solution). Thus, the reduction in the BSA sieving coefficient at low Cibacron Blue concentrations is likely due to the increased binding of Cibacron Blue, while the increase in the BSA sieving coefficient at high Cibacron Blue concentrations appears to be due to the increase in the concentration of free Cibacron Blue since this occurs after the protein is essentially saturated with bound Cibacron Blue.

In order to understand the complex behavior seen in Figure 3, experiments were performed at fixed ratios of Cibacron Blue to BSA for several values of solution ionic strength, which was adjusted by the addition of sodium chloride. Results are shown in Figure 4 for both the unmodified composite regenerated cellulose membrane (top panel) and the negatively charged version of the CRC membrane, with the data plotted as the mean plus/minus the standard deviation as determined from a minimum of three replicate experiments. The data were highly reproducible, with the BSA sieving coefficients typically varying by less than 7% among the replicates. The sieving coefficients obtained in the 100 mM solutions are nearly independent of the Cibacron Blue concentration, with values of approximately 0.6 for the neutral membrane and 0.12 for the negatively-charged membrane. The smaller sieving coefficients for the negatively-charged membrane at high salt are primarily due to the reduction in pore size associated with the chemical modification. The measured value of the hydraulic permeability of the charge-modified membrane was approximately 25% smaller than that for the native membrane, consistent with a reduction in effective pore size of about 0.8 nm as determined using the Hagen-Poiseuille equation. In contrast, the BSA sieving coefficient through the negatively-charged membrane in the 25 mM ionic strength solution decreased by more than two orders of magnitude upon addition of Cibacron Blue, going from 0.044 in the absence of Cibacron Blue to 0.0001 at a ratio of 12 molecules of Cibacron Blue to each molecule of BSA. A much smaller effect was seen with the unmodified CRC membrane, with

the sieving coefficient decreasing by less than a factor of two over this same range of Cibacron Blue concentrations.

The large reduction in protein transmission in the low ionic strength solution is due to the strong electrostatic exclusion of the negatively-charged complex formed when BSA binds large numbers of negatively-charged Cibacron Blue molecules. Model calculations for the net protein charge, accounting for the effects of charge regulation (see Appendix) indicate that the BSA charge decreases by approximately 13 electronic charges upon the addition of 12 moles of Cibacron Blue to the solution per mole of BSA. This accounts for the binding of 9.8 moles of Cibacron Blue per mole of BSA along with the change in ionization of the other amino acid residues caused by the preferential partitioning of H^+ into the immediate vicinity of the negatively-charged protein. This large increase in the net negative charge causes a very large electrostatic exclusion of the BSA from the pores of the negatively-charged membrane due to the direct charge-charge repulsion under these conditions. This effect is much less pronounced with the unmodified membrane, and it is almost completely eliminated at high ionic strength due to the strong electrostatic shielding provided by the bulk electrolyte under these conditions.

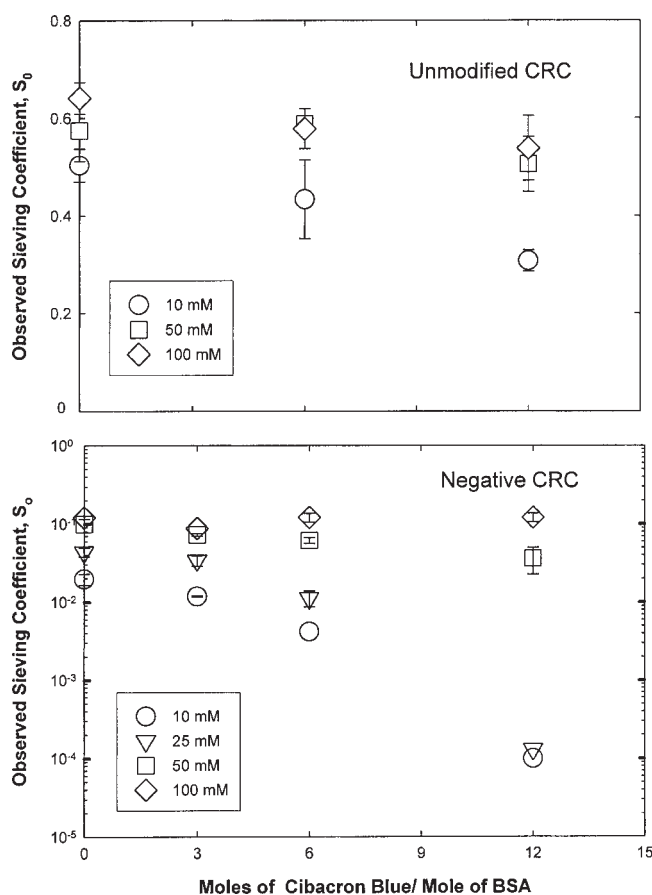


Figure 4. Observed sieving coefficients as a function of the molar ratio of Cibacron Blue to BSA at several ionic strength for an unmodified (top panel) and negatively charged (bottom panel) CRC membrane. All data were obtained at pH 5.0 using 8 g/L BSA.

Model Calculations

The BSA sieving coefficient data were analyzed using available hydrodynamic models accounting for the presence of a pore size distribution as (Zeman and Zydney, 1996):

$$S_a = \frac{\int_0^\infty \phi K_c n(r) r^4 dr}{\int_0^\infty n(r) r^4 dr} \quad (5)$$

where S_a is the actual sieving coefficient, which is equal to the ratio of the protein concentration in the filtrate solution to that in the solution immediately upstream of the membrane. ϕ is the equilibrium partition coefficient for the protein entering the pore and K_c is the hindrance factor for convection, which accounts for the additional hydrodynamic drag on the solute molecule in the presence of a pore wall. All calculations were performed using a log-normal pore density function, $n(r)$, where r is the pore radius (Zydney et al., 1994). The weighting by r^4 accounts for the dependence of the volumetric flow rate on pore radius.

The equilibrium partition coefficient for BSA was evaluated using the Smith and Deen (1980) model for the partitioning of a charged sphere in a charged cylindrical pore:

$$\phi = (1 - \lambda)^2 \exp\left(-\frac{\psi}{kT}\right) \quad (6)$$

where ψ/kT is the dimensionless electrostatic potential energy of interaction and λ is the ratio of solute to pore radii. The pre-exponential term accounts for the steric exclusion of the solute from the region within one solute radius of the pore wall. The electrostatic potential energy of interaction for a sphere at the pore axis was evaluated by solving the linearized Poisson–Boltzmann equation using matched expressions in spherical and cylindrical coordinates yielding (Smith and Deen, 1980):

$$\frac{\psi}{kT} = A_{ss}\sigma_s^2 + A_{sp}\sigma_s\sigma_p + A_{pp}\sigma_p^2 \quad (7)$$

where σ_s and σ_p are the surface charge density of the solute and pore wall, respectively, and A_{ss} , A_{sp} , and A_{pp} are all positive coefficients which depend on the solution ionic strength, pore radius, and solute radius. Expressions for these parameters have been summarized by Burns and Zydney (2001), although it should be noted that there is an error in their non-dimensionalization for the electrostatic energy of interaction (the energy was non-dimensionalized by $r_p \epsilon_0 \epsilon_r [RT/F]^2$ instead of kT as in Equation (7), where r_p is the membrane pore radius, ϵ_0 is the permittivity of free space, ϵ is the dielectric constant of the medium, and F is Faraday's constant).

As described by Equation (7), the electrostatic energy of interaction has three distinct contributions: one from the distortion of the electrical double layer surrounding the sphere (given by the term involving σ_s^2), one from direct

charge–charge interactions between the charged sphere and the membrane pores, and one that accounts for the deformation of the electrical double layer within the pore. These phenomena have been discussed in more detail by Burns and Zydney (2001).

Model calculations were performed as follows. First, the extent of Cibacron Blue binding to BSA was evaluated using the previously determined binding isotherm (Eq. 4). The net charge on the BSA–Cibacron Blue complex was then evaluated using the charge regulation model described in Appendix A which accounts for the dissociation of the acidic and basic amino acids on the protein as well as the binding of acetate anions (Eqs. A1 to A6). The solution ionic strength was then determined as:

$$I = \frac{1}{2} \sum_i z_i^2 C_i \quad (8)$$

where the summation is over all (free) ionic species with z_i the ion valence and C_i the concentration of the free ion. Eq. (8) thus accounts for the presence of acetate, Na^+ , Cl^- , and the negatively-charged Cibacron Blue, with the concentration of the latter determined from the free Cibacron Blue concentration (evaluated from the binding isotherm) with $z_i = 3$. The surface charge density of the unmodified cellulose membrane was $\sigma_p = -0.0002 \text{ C/m}^2$ while that for the negatively-charged membrane was $\sigma_p = -0.01 \text{ C/m}^2$, with both of these values determined from streaming potential measurements as discussed by Burns and Zydney (2000). The electrostatic energy of interaction was then calculated using Equation (7), with the actual sieving coefficient determined by integration over the log-normal distribution using a mean pore radius of 62 Å for the unmodified membrane and 54 Å for the negatively-charged membrane with $\sigma/r_{\text{mean}} = 0.2$ in both cases. The choice of $\sigma/r_{\text{mean}} = 0.2$ was based on previous studies of similar ultrafiltration membranes (Zeman and Zydney, 1996). The mean pore radius was evaluated from the experimentally measured values of the hydraulic permeability ($L_p = 3.0 \times 10^{-12} \text{ m}$ for an unmodified membrane and $L_p = 2.3 \times 10^{-12} \text{ m}$ for a charge modified version) using a membrane porosity of 50% and skin thickness of 1 µm based on the analysis presented in Zeman and Zydney (1996).

Figure 5 shows a comparison between the model calculations and the experimental data for the actual protein sieving coefficient, S_a , as a function of the Cibacron Blue concentration for the 10 mM ionic strength solution using both the unmodified and negatively-charged membranes. The actual sieving coefficients were calculated from the measured values of the observed sieving coefficients using a simple stagnant film model to account for the effects of concentration polarization in the stirred cell (Zeman and Zydney, 1996):

$$S_a = \frac{S_0}{(1 - S_0) \exp\left(\frac{J_v}{k}\right) + S_0} \quad (9)$$

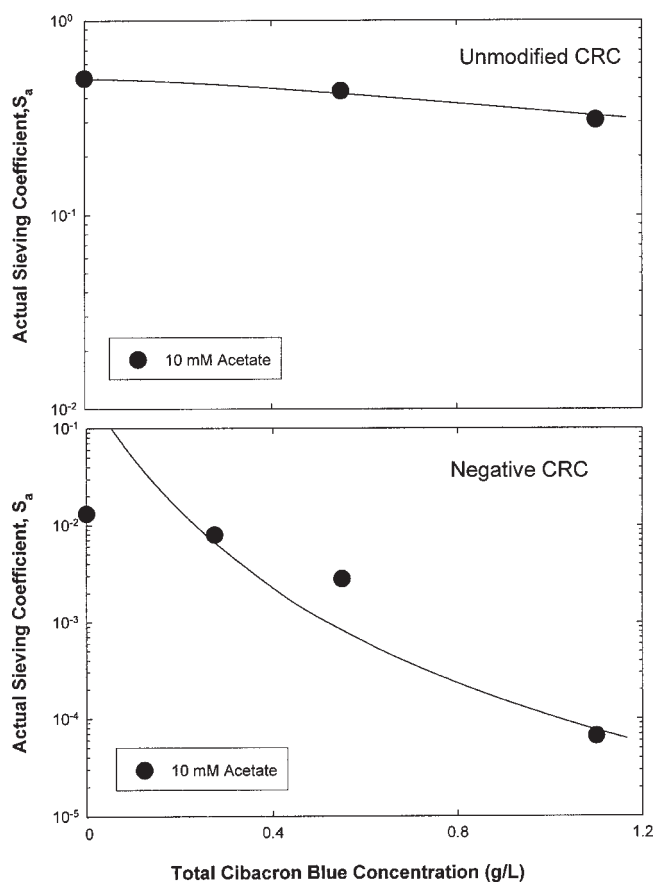


Figure 5. Actual sieving coefficients for BSA through the unmodified (top panel) and negatively charged (bottom panel) 100 kDa CRC membrane as a function of the total Cibacron Blue concentration in a 10 mM acetate buffer at pH 5.0. Solid curves represent model calculations as described in the text.

where J_v is the filtrate flux and k is the bulk mass transfer coefficient, which was estimated as $9.6 \mu\text{m/s}$ under the conditions of these experiments (Pujar and Zydney, 1994). The model calculations are in good agreement with the data for both unmodified and negatively-charged versions of the membrane over the entire range of Cibacron Blue concentrations. The model does tend to over-predict the actual sieving coefficient for the negatively-charged membrane in the absence of Cibacron Blue, which is due to the small positive charge that is calculated for BSA under these conditions using the charge regulation model presented in Appendix A. If the BSA charge were reduced by only a single electronic charge, the calculated value for S_a would be in excellent agreement with the data at zero Cibacron Blue concentration. Such small deviations in the BSA charge for different protein lots can arise from the presence of different isoforms of BSA (e.g., different intermolecular disulfide linkages) and different amounts of fatty acid.

The effect of the Cibacron Blue concentration on the sieving behavior is examined in more detail in Figure 6. The upper panel shows the calculated values of the net protein charge on the left axis and the calculated values of the solution ionic strength on the right axis, both as a function of the total Cibacron Blue concentration added to

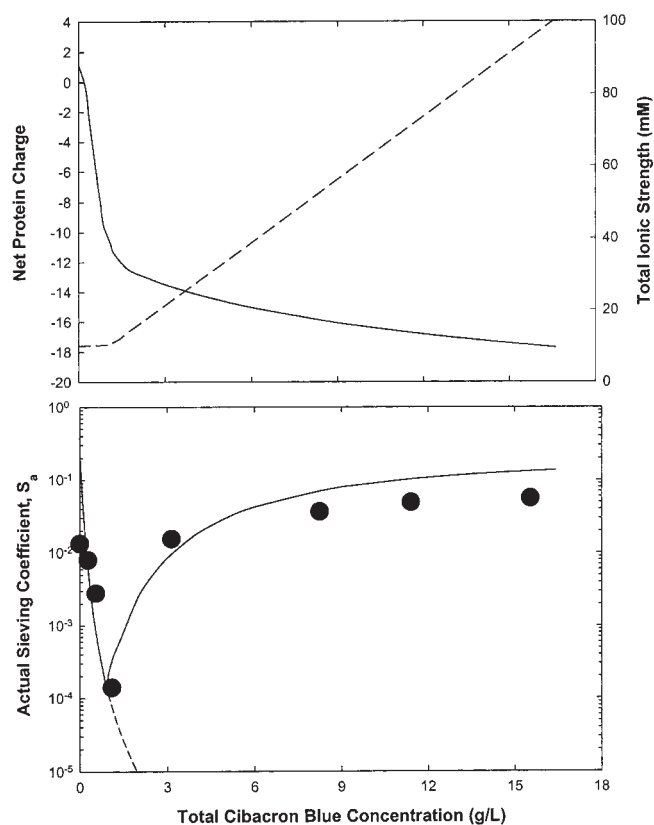


Figure 6. Effect of Cibacron Blue on the actual sieving coefficient for BSA in a 10 mM acetate buffer at pH 5 using an 8 g/L BSA solution. The top panel shows the net protein charge (solid line) and overall solution ionic strength (dashed line). The bottom panel shows the actual sieving coefficient, with the solid curve evaluated using the full model and the dashed curve evaluated assuming a constant ionic strength of 10 mM.

the protein solution. Model calculations for the net protein charge were performed using a BSA concentration of 8 g/L at pH 5.0 in the presence of 10 mM acetate buffer. The addition of Cibacron Blue causes a rapid increase in the net negative charge of BSA until the total Cibacron Blue concentration exceeds about 1.0 g/L, after which the BSA charge decreases relatively slowly with increasing amounts of Cibacron Blue. The rapid initial reduction in protein charge is due to Cibacron Blue binding, while the weaker variation at high Cibacron Blue concentrations is due to the change in protonation of the acidic and basic amino acids caused by the reduction in H^+ concentration near the surface of the protein due to the increased electrostatic shielding caused by the high ionic strength. The solution ionic strength is a very weak function of the Cibacron Blue concentration at low concentrations of Cibacron Blue since most of the dye is bound by the protein and thus provides no contribution to the solution ionic strength. The ionic strength begins to increase linearly with the Cibacron Blue concentration for Cibacron Blue concentrations greater than 1.0 g/L since the protein is fully saturated with the dye under these conditions.

The lower panel shows the calculated values of the actual sieving coefficient as a function of the total concentration of Cibacron Blue. The solid symbols are the experimental

results, with the S_a values calculated from the observed sieving coefficient data in Figure 3 using the stagnant film model (Eq. 9). The solid curve is the full model calculation, which properly captures both the reduction in BSA sieving coefficient at low Cibacron Blue concentrations and the minimum at a Cibacron Blue concentration of 1.0 g/L. The dashed curve shows the model calculations accounting for the change in BSA charge upon the addition of Cibacron Blue but neglecting any change in the solution ionic strength (which was kept constant at 10 mM). The dashed curve accurately describes the data at low Cibacron Blue concentrations, confirming that the reduction in the sieving coefficient under these conditions is a direct result of the change in net protein charge associated with the binding of the negatively-charged Cibacron Blue molecules. The dashed curve dramatically underpredicts the sieving coefficient at higher Cibacron Blue concentrations since it neglects the increase in ionic strength, and thus the reduction in electrostatic repulsion, that occurs under these conditions. The full model, which accounts for both the change in protein charge and the change in solution ionic strength (as shown in the top panel), is in excellent agreement with the experimental data over the full range of Cibacron Blue concentrations.

DISCUSSION

The experimental data obtained in this study clearly demonstrate the feasibility of using a small charged ligand to control protein transport during ultrafiltration. In this case, the addition of approximately 1 g/L of Cibacron Blue to an 8 g/L solution of BSA caused a reduction in the BSA sieving coefficient by more than two orders of magnitude, from 0.02 to approximately 10^{-4} , due to the strong electrostatic exclusion of the negatively-charged BSA complex from the negatively-charged membrane. This effect is completely eliminated at high ionic strengths and is reduced when using a neutral version of the CRC membrane due to the reduction in the electrostatic repulsion.

The effect of Cibacron Blue on the BSA sieving coefficient arises from two distinct phenomena: the change in net protein charge due to binding of the negatively-charged dye and the change in solution ionic strength due to the presence of free ligand in solution. The first effect dominates at low Cibacron Blue concentrations since most of the added dye binds to the protein. The second effect dominates at high Cibacron Blue concentrations where the binding sites on the protein are fully saturated with dye. Model calculations accounting for both of these phenomena are in excellent agreement with the experimental data.

The presence of these two contributions leads to a distinct optimum in the ligand concentration for maximizing the electrostatic interactions and in turn maximizing the extent of protein retention. For the BSA–Cibacron Blue system, this optimal ligand concentration corresponds to conditions in which there is just sufficient ligand to fully saturate all of the binding sites on the protein. Since BSA can bind up to

10 molecules of Cibacron Blue, the increase in electrostatic interactions is quite dramatic, leading to a two order of magnitude reduction in protein transmission through the negatively-charged membrane.

In order for this approach to be used commercially, it would be necessary to design a process to dissociate and then separate the protein and dye, with the dye recycled for subsequent processes. In the case of Cibacron Blue and BSA, it is known that the binding interactions become negligible at very high pH and in the presence of salts like NaSCN. For example, Ruckenstein and Zeng (1998) showed that 99% of BSA can be recovered from a Cibacron Blue affinity membrane using a 0.5 M solution of NaSCN in a 0.05 M tris buffer at pH 8. Free Cibacron Blue could be recovered by diafiltration using this buffer, with the BSA retained by using a very small pore size (e.g., 10 kDa nominal molecular weight cut-off) CRC membrane.

Although all of the data obtained in this study were with a single protein, it should also be possible to exploit the specificity of the dye-binding interactions to enhance the selectivity of protein separations. In this case, one would look to use a dye that specifically binds to the product of interest, significantly increasing its negative charge and thus its retention during high performance tangential flow filtration. In principle, this approach could be used to separate proteins with nearly identical surface charge (or isoelectric points) using highly charged biospecific dye ligands. The pure proteins would then be recovered by a subsequent diafiltration performed using a buffer that causes the dissociation of the protein–dye complexes. This could provide new opportunities for using membrane systems for protein separations.

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Appendix

The net charge of BSA was evaluated using the charge regulation model originally developed by Tanford et al. (1955) and Scatchard et al. (1957), but modified to account for the binding of Cibacron Blue. The net charge was evaluated from the difference in the number of protonated amino acids, the number of bound acetate anions, and the number of bound molecules of Cibacron Blue:

$$Z_{BSA} = v_{H^+} - v_{acetate} - 3v_{CiB} \quad (A1)$$

where each bound Cibacron Blue has a charge of -3 .

The number of positively charged amino acid residues is calculated by summing over seven different types of residues:

$$v_{H^+} = 96 - \sum_i \frac{n_i K_i^{int}}{K_i^{int} + [H_b^+]} \exp(-e\psi_s^*/kT) \quad (A2)$$

where $[H_b^+]$ is the bulk hydrogen ion concentration, and the number (n_i) and intrinsic equilibrium constants (K_i^{int}) for

Table AI. Type and number of titratable groups on BSA.

Type (<i>i</i>)	Number (n_i)	pK_{int}^i
α -Carboxyl	1	3.75
β , γ -Carboxyl	99	4.02
Imidazole	16	6.9
α -Amino	1	7.75
ϵ -Amino	57	9.8
Phenolic	19	10.35
Guanidine	22	12

each site are given in Table AI. The total number of positively charged amino acid residues at very low pH, i.e., where all the sites are protonated, is +96. The exponential term in Eq. (A2) is the Boltzmann factor and accounts for the partitioning of the hydrogen ions into the region immediately adjacent to the protein surface due to electrostatic interactions. Thus, the H^+ concentration near a negatively-charged protein will be greater than the bulk value, with the reverse being true for a positively-charged protein. ψ_s^* is the electrostatic potential at the protein surface and is related to the net protein charge as (Pujar and Zydney, 1997):

$$\sigma_s^* = \frac{eZ_{complex}}{4\pi r_s^2} = \frac{\epsilon_0 \epsilon \psi_s^* (1 + \kappa r_s)}{r_s} \quad (A3)$$

where e is the electron charge, r_s is the radius of the (spherical) BSA, ϵ_0 is the permittivity of free space, ϵ is the dielectric constant of the medium, and κ is the inverse of the Debye length:

$$\kappa^{-1} = \left(\frac{\epsilon_0 \epsilon RT}{F^2 \sum z_i^2 c_i} \right)^{1/2} \quad (A4)$$

The number of bound acetate ions (CH_3COO^-) was evaluated similarly as

$$v_{CH_3COO^-} = \sum_j \frac{m_j K_j \gamma [CH_3COO^-] \exp(e\psi_s^*/kT)}{1 + K_j \gamma [CH_3COO^-] \exp(e\psi_s^*/kT)} \quad (A5)$$

The parameters m_j and K_j for the three distinct acetate binding sites are given in Table AII. γ is the activity coefficient of CH_3COO^- and is evaluated as (Scatchard et al., 1957)

$$-\log \gamma = \frac{0.5 \sqrt{[CH_3COO^-]/2}}{(1 + 2 \sqrt{[CH_3COO^-]/2})} \quad (A6)$$

Calculations were performed for the 10 mM acetate solution so there was no need to account for any chloride ion binding.

The net protein charge in a given solution is evaluated by simultaneously solving Eqs. A1 to A6 with the number of

Table AII. Values of parameters m_j and K_j in the acetate binding model.

j	m_j	K_j (L/mol)
1	1	530
2	6	50
3	14	2

bound molecules of Cibacron Blue given by Eq. (4). The radius of BSA was fixed at $r_s = 3.48$ nm (Axelsson, 1978) independent of the number of molecules of bound Cibacron Blue since Cibacron Blue is so much smaller than BSA. The net charge of BSA in the 10 mM acetate buffer is calculated as +1.2 while the charge in the presence of 1.0 g/L Cibacron Blue is −11.8 (for a BSA concentration of 8 g/L).

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